General Science

Alveary

About the Course

Learners engage with the chemical nature of Things, meteorology, space exploration, and robotics as they practice beginning lab skills and expand their use of graphs. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 5

About Science: Grade 5
In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

Science: Grade 5
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 5
In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

General Science: Grade 5
For fifth-grade students or possibly hungry fourth-graders or sixth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
5	General Science: Grade 5 1 time/week 20 min	Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt The Hive Detectives: Chronicle of a Honey Bee Catastrophe The Tornado Scientist

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 5				
Natural History: Grade 5		General Science: Grade 5 Nature Walks & Scouting: Grades 1-8		Labs: Grade 5 Nature Notebook: Grade 5

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 5
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 5
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 5
☐ Term 1: a beekeeper
☐ Term 2: a meteorologist or weather station
☐ Term 3: evening walks, a planetarium

Term Prep & Teacher Tips

Science: Grade 5
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 5
☐ Term 1: bees p.384-395
☐ Term 2: climate and weather p.780-783, the winds and storms p.791-799
☐ Term 3: stars p.815-821

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- honey
- food items suggested
- spoons
- small saucepan
- plant fertilizer of your choice (only a small amount)
- water
- flowers (Term 1)
- a strainer or sieve
- 1 c dry soil
- kitchen scale
- kitchen thermometer, any stick-type tool that can be placed into hot water and soil
- oven
- oven mitt
- clock
- chair or stepladder
- stopwatch, such as found on most electronic devices
- freezer
- stove
- 1-2 c white vinegar
- 1/2-1 c baking soda
- flat surface outside to launch rocket
- 10-20' height reference, such as the side of a building, a long piece of wood, or a flagpole
- various supplies by student design
- printables
- optional: barometer
- optional: computer

Reminders

General Science: Grade 5
☐ Note that Lab in week 1 begins with 5 "unknown" food substances (1- 2 Tbsp) to include honey and any other various items, e.g. peanut butter, whole grain bread, white bread, raw egg white, cereal, applesauce, jelly, whole milk, milk substitute, juice, flavored yogurt, plain yogurt, etc. These will be provided to students in 5 similar-looking containers for study. Refer to the lab book for more details.

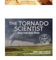
Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

General Science: Grade 5


Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt


The Hive Detectives: Chronicle of a Honey Bee Catastrophe


The Tornado Scientist

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 5

Household Items - General Science: Grade 5

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

General Science: Grade 5

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

General Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

☐ **20m General Science: Grade 5 - Lesson 1**
Visit a Hive

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
Look at the cover picture and the main title. Have you watched bees before? What did you notice about them? What would you investigate if you were a "hive detective"? Read the subtitle. Do you know what a chronicle is? (a factual record) Do you know what a catastrophe is? (a great disaster) So, this is a story about scientists who are studying a disaster in a beehive to learn how it happened in the hopes that they can prevent it from happening again.

→ READ, NARRATE, & DISCUSS
p.1-3 "Put on your" - "home in Wisconsin."

→ SUPPLEMENTAL
∞ Video Link: Why do Honeybees Love Hexagons? (3:58)
∞ Video Link: Why Do Bees Build Hexagons? (2:34)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 2

☐ **20m General Science: Grade 5 - Lesson 2**
Know the Bees

Materials: The Hive Detectives

→ INTRO
What did you read last time?

→ READ, NARRATE, & DISCUSS
p.4-9 "Mary begins" - "for their keeper."

● Describe the structure of the hive and the jobs of the bees. Draw or diagram, if helpful.

WEEK 3

☐ **20m General Science: Grade 5 - Lesson 3**
Beekeeping Cross-Country

Materials: The Hive Detectives

→ INTRO
What did you read last time? Some beekeepers manage thousands of bees. Who would need thousands of bees? Let's find out about what happened when one of these beekeepers visited his hives.

→ READ, NARRATE, & DISCUSS
p.11-14 "On November 12," - "a thankless job."

● Contrast how Mary and Dave keep their bees.
● What would it feel like to come upon the animals you care for to find them gone?

→ SUPPLEMENTAL
∞ Video Link: Honey Bee Shortage (2:34)

General Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 4

20m General Science: Grade 5 - Lesson 4
Colony Collapse Disorder

Materials: The Hive Detectives

→ INTRO
What did you read last time? How do you think other beekeepers and scientists reacted when they heard what happened to Dave's bees?

→ READ, NARRATE, & DISCUSS
p.15-17 "In all my" - "and stop it."

• How do you think the scientists might test their three theories?

→ SUPPLEMENTAL
∞ Video Link: How Do Bees Make Honey? (to timestamp 3:20)

→ NOTE
Read A Dream Team of Bee Scientists p.18-19, if desired.

• FIELD TRIP
Visit a beekeeper. Notice the bees come and go from the hive. Notice how they interact with each other at the entrance. Notice their behavior when the beekeeper opens the hive. Can you find the queen? How does she look different? Where is she in the hive? Are there larvae in the hive? Where are they?

WEEK 5

20m General Science: Grade 5 - Lesson 5
Learning from Bees

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
What did you read last time? The first step to solving a problem is to learn more, so let's see how they do that today.

→ READ, NARRATE, & DISCUSS
p.20-25 "The CCD Working Group" - "But what?"

• Describe the samples scientists collected.
• What did the scientists learn so far from studying the bees?
• How does a bee pollinate flowers? It's actually pretty amazing! Watch the videos to take a closer look:
∞ Video Link: The Buzz About Bees (2:46)
∞ Video Link: Sweet Pea Pollination (2:13)

→ NOTE
Read Inside the Hive p.26-27, if desired.

★ TEACHER NOTE
The second supplemental video mentions evolution at 0:24. Teachers might choose to introduce the term by the way, mute the narration, or skip this one.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in bees? More interest/ease in describing or thinking about what material Things are made of? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve, bee keeper, or museum with a chemistry exhibit. Reach out through Contact Us, if you need help.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 6

20m General Science: Grade 5 - Lesson 6
The Pest Hypothesis

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
What did you read last time? One hypothesis is that a known pest is killing the bees. What's a hypothesis? (a really good guess based on what we know) To find out if the guess that a known pest is killing the bees is accurate, they have to test it. Let's find out how they do that today.

→ READ, NARRATE, & DISCUSS
p.29-31 "When asked" - "Strike three."

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

Term 1

→ **SUPPLEMENTAL**
∞ Video Link: How Do Bees Make Honey? (timestamp 3:20 to end)

→ **NOTE**
Read Inside the Hive p.32-33, if desired.

WEEK 7

20m General Science: Grade 5 - Lesson 7
The Virus Hypothesis

Materials: The Hive Detectives

→ **INTRO**
What did you read last time? Another hypothesis is that a virus might be making the bees sick, but viruses can't be seen with our eyes or even a microscope. How do you think they will test this hypothesis?

→ **READ, NARRATE, & DISCUSS**
p.34-37 "Diana Cox-Foster" - "raised indoors?"

- How might Diana design her study so that she can find out what she needs to know without making the CCD problem worse?
- What do you think—can honeybees be raised indoors?

WEEK 8

20m General Science: Grade 5 - Lesson 8
Thinking about Bee Health

Materials: The Hive Detectives

→ **INTRO**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
p.37-39 "Well, not yet." - "a greater issue?"

→ **NOTE**
Read Bee Bodies p.40-41, if desired.

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 9

20m General Science: Grade 5 - Lesson 9
Caring for Creation

Materials: The Hive Detectives

→ **INTRO**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
p.43-47 "One year after" - "knows about bees."

- Tell what scientists now know about CCD and its causes.
- How does our knowing bees make a difference?

→ **SUPPLEMENTAL**
∞ Video Link: Can Mushrooms Save the Honeybee? (5:45)

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 10

20m General Science: Grade 5 - Lesson 10
Being Cared for By Creation

Materials: The Hive Detectives

→ **INTRO**
What did you read last time?

Term 1

→ READ, NARRATE, & DISCUSS
p.49-52 "It's the middle" - "several local fairs.)"

• Let's think of some of the ways that people's lives are better because of the bees. Some ways might have been mentioned in the book, or might have nothing to do with the book.
• Make up a poem about bees together!

→ SUPPLEMENTAL
∞ Video Link: How to Plant a Pollinator Garden (3:21)
∞ Video Link: How to Harvest Honey (8:47)

WEEK 11

⬚ **20m General Science: Grade 5 - Lesson 11**
Final Thoughts

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
What did you read last time?

→ READ, NARRATE, & DISCUSS
p.53-55 "Mary repeats" - "see them through."

→ SUPPLEMENTAL
A master beekeeper shares some important thoughts about biodiversity in the linked article. Teachers might share some of those thoughts along with the video. What are some ways that we can help all the bees?
∞ Article Link: Honeybee Suite
∞ Video Link: This Bee Works 50 Times Harder (1:55)

→ NOTE
Read any parts of Appendix Bee p.56-59, if desired.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether bees, our dependence on them, or what material Things are made of are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 12

⬚ **20m General Science: Grade 5 - Lesson 12**
Exam Week

→ EXAMS
Answer question(s) related to course.

Questions will come from:
The Hive Detectives

General Science: Grade 5

Click THIS text or scan the QR code for links.

Term 2

WEEK 1

⬚ **20m General Science: Grade 5 - Lesson 13**
Who What Where

Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO
What kinds of storms do you experience in your region? Many people have tornadoes, and this book is about how scientists are trying to better understand when and where these storms will occur. Why do you think that would be important? Our story takes place in the United States, but tornadoes don't pay attention to political boundaries! Canada is one of the most tornado-prone countries, and tornadoes occur all over the world.

→ READ, NARRATE, & DISCUSS
p.1-5 "Beep... beep...beep" - "ever wanted to be."

• Draw or tell about a storm that you remember.

★ TEACHER TIP
Severe weather can be scary for some students, and we want to honor their experiences. Notice how they respond to the story and respond with sensitivity and encouragement. Some will find the information helps them overcome their fears, while others may need more time.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2

⬚ **20m General Science: Grade 5 - Lesson 14**
Robin's Story

Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO
Recall what we read about in the last passage. Today, let's find out how Robin became a tornado scientist.

→ READ, NARRATE, & DISCUSS
p.6-11 "Robin! Come and" - "family history, too."

→ SUPPLEMENTAL
∞ Video Link: 1986 Tornado Footage (4:58)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 3

⬚ **20m General Science: Grade 5 - Lesson 15**
Twister Science

Materials: The Tornado Scientist

→ INTRO
Tell me what you remember from the last reading. Today, Robin is going to teach us about how a tornado works and how scientists learn about them.

→ READ, NARRATE, & DISCUSS
p.13-17 "Look at that!" - "of cold air."

→ SUPPLEMENTAL
∞ Video Link: What Causes a Tornado? (3:05)

WEEK 4

⬚ **20m General Science: Grade 5 - Lesson 16**
Life Cycle of a Storm

Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO
What do you remember about tornadoes from the last lesson? Let's hear more about the life cycle of these storms.

★ TEACHER TIP
Some students may benefit from reading and narrating this lesson in parts, as presented here. Help them notice if there are parts to the reading; these can help them organize the ideas in their minds.

→ **READ & NARRATE**
p.18-19 "You probably know" - "spinning funnel."

→ **READ, NARRATE, & DISCUSS**
p.20-23 "Tornadoes are a" - "tornado chases you."

→ **SUPPLEMENTAL**
Today's reading discusses storm formation. Do you remember in what part of the atmosphere this happens? Check out the images at the following website. Can you find where weather occurs? Where weather balloons rise to? Where aurora borealis occurs?
∞ Website Link: Layers of the Atmosphere

WEEK 5

20m General Science: Grade 5 - Lesson 17
Seeing Storms

Materials: The Tornado Scientist

→ **INTRO**
Recall what you've read so far about storms. How do you think scientists see inside storms?

→ **READ, NARRATE, & DISCUSS**
p.24-29 "Spotting a tornado" - "storm is moving."

★ TEACHER NOTE
The section subtitled "Surviving the Greensburg Tornado" p.26-27 provides a detailed experience of a storm. Teachers should preread if they have any concerns. They might choose to skip this section or allow students to help decide if they want to read it.

WEEK 6

20m General Science: Grade 5 - Lesson 18
Weather Experiments

Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ **INTRO**
Recall from the last reading how scientists "see" storms. What do they do with their observations?

→ **READ, NARRATE, & DISCUSS**
p.30-33 "My job is" - "target? Dixie Alley."

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in weather? More interest/ease in collecting weather data or thinking about the instruments used to do so? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a weather station. Reach out through Contact Us, if you need help.

WEEK 7

20m General Science: Grade 5 - Lesson 19
Still Learning

Materials: The Tornado Scientist

→ **INTRO**
Remind me what you've learned about how scientists "see" storms and what they do with those observations. So, do you think they've got tornadoes figured out?

→ **READ, NARRATE, & DISCUSS**
p.35-41 "Eighth-grader Andrew" - "stop: Huntsville, Alabama."

- Describe the characteristics that make southern tornadoes and plains tornadoes different.

★ TEACHER NOTE
The first part of this reading p.35-36 provides a detailed experience of a storm. Teachers should preread if they have any concerns. They might choose to begin the reading from the subtitle "Southern Shocker" p.36 or allow students to help decide if they want to read it.

General Science: Grade 5

Click THIS text or scan the QR code for links.

Term 2

WEEK 8	⬚ **20m General Science: Grade 5 - Lesson 20** *The Tornado Trap* ▯ Materials: The Tornado Scientist **→ INTRO** Recall what is happening with Robin and her fellow weather scientists. What kinds of questions do you think they should ask about storms? How can they collect better observations? **→ READ, NARRATE, & DISCUSS** p.42-51 "Where do scientists" - "goes into SASSI." • What information have you been recording on your weather chart? How is the data collected by the scientists in your book similar or different? Take a closer look at how they collect this data in their tornado trap: ∞ Video Link: Storm Pods (1:34)	
WEEK 9	⬚ **20m General Science: Grade 5 - Lesson 21** *Problem-Solving* ▯ Materials: The Tornado Scientist PREP: Read Teacher Tip **→ INTRO** Tell me how the tornado research project is going so far. Let's find out how Robin is going to problem-solve. **→ READ, NARRATE, & DISCUSS** p.52-57 "Back in Alabama" - "Bingo. It's working." **→ SUPPLEMENTAL** Is the humble weather balloon still important? ∞ Video Link: Weather Balloons (2:43)	★ TEACHER TIP Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s). **• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 10	⬚ **20m General Science: Grade 5 - Lesson 22** *Reasons for Research* ▯ Materials: The Tornado Scientist **→ INTRO** Remember back to the beginning of the book. Why do scientists want to learn more about tornadoes? **→ READ, NARRATE, & DISCUSS** p.58-65 "The weather seems" - "ever wanted to be." • Diagram how tornadoes form. **→ SUPPLEMENTAL** ∞ Video Link: Hello Kitty Goes to Space (3:46)	**• FIELD TRIP** Visit a meteorologist or weather station.
WEEK 11	⬚ **20m General Science: Grade 5 - Lesson 23** *Catch Up* PREP: Read Teacher Tip **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When student(s) take exams next week, evaluate what they recall, but more importantly, whether tornadoes, weather, or instrumentation are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider

[Click THIS text or scan the QR code for links.]

Term 2

		why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.
WEEK 12	⬚ **20m General Science: Grade 5 - Lesson 24** *Exam Week* **➜ EXAMS** Answer question(s) related to course. Questions will come from: The Tornado Scientist	

General Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 1

⬚ **20m General Science: Grade 5 - Lesson 25**

The Launch

Materials: Mission to Pluto

PREP: Read Teacher Tip

➔ INTRO
Pluto is at the edge of our solar system. This is a story about scientists who study this distant object. Why do you suppose they would want to do that?

➔ READ, NARRATE, & DISCUSS
p.1-6, 10 "The large meeting" - "So stay tuned."

➔ SUPPLEMENTAL
∞ Video Link: New Horizons Launch (1:35)

➔ NOTE
Explore The Journey to Pluto p.8-9 for interest, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 2

⬚ **20m General Science: Grade 5 - Lesson 26**

The Path to a Space Mission

Materials: Mission to Pluto

➔ INTRO
Recall what the launch party was like and why people were so excited. Today let's find out how we got to the launch.

➔ READ, NARRATE, & DISCUSS
p.11-17, 20 "Three years after" - "pad was on."

➔ NOTE
Explore any sidebars and Pluto Primer p.18-19 for interest, if desired.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3

⬚ **20m General Science: Grade 5 - Lesson 27**

Building a Spacecraft

Materials: Mission to Pluto

PREP: Read Teacher Tip

➔ INTRO
Recall how Pluto was discovered, and eventually, scientists decided to study it further. To accomplish their mission, they will need a spacecraft. What do you think are important characteristics of a spacecraft?

➔ READ, NARRATE, & DISCUSS
p.21-25, 28 "Outer space may" - "engineers have done."

➔ NOTE
Explore any sidebars and The Spacecraft p.26-27 for interest, if desired.

★ TEACHER TIP
This passage describes the design of New Horizons. Therefore, the narration is more likely to be points of that description.

WEEK 4

⬚ **20m General Science: Grade 5 - Lesson 28**

Collecting Data

Materials: Mission to Pluto

PREP: Read Teacher Tip

➔ INTRO

★ TEACHER TIP
The first part of this passage describes the instruments on New Horizons (and the corresponding data to be collected), while the second narrates launch day from the scientists' perspective.

Tell me what was important about the design of New Horizons. Today we'll read about the kinds of information they want to collect with New Horizons and what launch day was like for the scientists who designed it.

→ READ & NARRATE
p.28-32 "Looking for ice" - "It's a great job."

→ READ, NARRATE, & DISCUSS
p.32-34 "At last" - "it got there."

→ NOTE
Explore any sidebars and Seeing Across the Spectrum p.31 for interest, if desired.

WEEK 5

⬚ **20m General Science: Grade 5 - Lesson 29**
The Path to Pluto

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO
What do you remember about New Horizons? What do you think it was like being one of the scientists who designed the spacecraft? How do you think you fly a spacecraft across space?

→ READ, NARRATE, & DISCUSS
p.35-39 "The Mission" - "said Alan."

→ SUPPLEMENTAL
∞ Video Link: Tour of the Galaxy (2:59)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

● NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 6

⬚ **20m General Science: Grade 5 - Lesson 30**
Planning for the Unexpected

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO
Tell me about flying a spacecraft across space. What kinds of surprises do you think might come up on such a journey?

→ READ, NARRATE, & DISCUSS
p.39, 41-42, 46 "Practice makes perfect" - "Stay tuned."

→ SUPPLEMENTAL
∞ Video Link: Relive the Moon Landing (2:15)

→ NOTE
Explore Minimoons p.44-45 for interest, if desired.

★ TEACHER TIP
This is another description of points passage!

★ TEACHER NOTE
The supplemental reading "Minimoons" includes references to early planet/moon formation. Teachers might choose to allow it to pass by the way, take the opportunity to discuss other ideas, or skip this one.

WEEK 7

⬚ **20m General Science: Grade 5 - Lesson 31**
What is a Planet Anyway?

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO
What do you recall about what it's like to fly a spacecraft?

→ READ, NARRATE, & DISCUSS

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in the solar system? More interest/ease in observing outer space or the technology used to do so? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us, if you need help.

p.40-41 sidebar "As New Horizons" - "that Pluto fails."

● It seems that we humans like to classify and group Things. What groups do we use for planets? What other Things do you know that we classify into groups? Animals? Plants? Rocks?
● Act out a news flash to tell people why Pluto is no longer classified as a planet.

WEEK 8

20m General Science: Grade 5 - Lesson 32
Hello from Pluto!

Materials: Mission to Pluto

→ INTRO
Remind me: Is Pluto a planet or not? Why or why not? Let's continue there with New Horizons.

→ READ, NARRATE, & DISCUSS
p.47-49, 52-53 "It had already" - "are studied."

→ NOTE
Explore Pluto Encounter p.50-51 for interest, if desired.

★ TEACHER NOTE
The age of the Earth is mentioned in the first full paragraph on p.53: "for billions of years." Teachers might choose to allow it to pass by the way, decide how they would like to discuss with students, or substitute "for a very long time" if reading aloud.

WEEK 9

20m General Science: Grade 5 - Lesson 33
Pluto's Secrets

Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO
Tell me what happened in the last reading on the day of the flyby. Let's find out about some of the first insights that New Horizons has shared.

→ READ, NARRATE, & DISCUSS
p.53-57, 60 "An active world" - "incredible beauty."

● Make up a poem about Pluto together!

→ SUPPLEMENTAL
∞ Resource Link: NASA New Horizons

→ NOTE
Explore New Horizons' Discoveries So Far p.58-59 for interest, if desired.

★ TEACHER TIP
This is another description of points passage!

★ TEACHER NOTE
The early formation of the Earth is discussed in the last paragraph on p.54: "Earth's surface... newborn planet." Teachers might choose to let it pass by the way or consider alternative theories they would like to present.

WEEK 10

20m General Science: Grade 5 - Lesson 34
Meet the Neighbors

Materials: Mission to Pluto

→ INTRO
Recall some of the discoveries made by New Horizons. So, there is this spacecraft flying out at the edge of our solar system. What do you think happens to it now that it has made it to Pluto?

→ READ, NARRATE, & DISCUSS
p.61-63 "What's next" - "go find them."

★ TEACHER NOTE
The early formation of the solar system is discussed in the first full paragraph on p.62: "KBOs also hold... John Spencer." Teachers might choose to let it pass by the way or consider alternative theories they would like to present.

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[Click THIS text or scan the QR code for links.]

Term 3

WEEK 11	⬚ **20m General Science: Grade 5 - Lesson 35** *Going and going...* Materials: Mission to Pluto PREP: Read Teacher Tip **→ INTRO** Recall what the plan is for New Horizons now that it has made its observations at Pluto. Let's finish the story today. **→ READ, NARRATE, & DISCUSS** p.63-65, 68 "Astronomers put" - "-five or so years." **→ SUPPLEMENTAL** ∞ Video Link: New Horizons Tribute (2:35) **→ NOTE** Explore The Kuiper Belt p.66-67 for interest, if desired.	★ TEACHER TIP When student(s) take exams next week, evaluate what they recall, but more importantly, whether the solar system, stars, or related technology are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help. **● NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 12	⬚ **20m General Science: Grade 5 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Mission to Pluto	

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Examination

Term 1

· The Hive Detectives: How would a chemist answer the question “What is honey?” OR Give possible reasons that honey bee populations are at risk, and explain what you can do to help.

· Explain how you did one investigation in science lab this term and what you learned from it.

Term 2

· The Tornado Scientist: Tell about or draw how meteorologists use instruments to conduct field work, like studying tornadoes. OR Explain what impact scientists hope to have by studying severe weather events, like tornadoes.

· Explain how you did one investigation in science lab this term and what you learned from it.

Term 3

· Mission to Pluto: Tell about or draw one way that scientists learn about our universe. OR Explain why scientists study far away Things, like Pluto.

· Explain how you did one investigation in science lab this term and what you learned from it.